

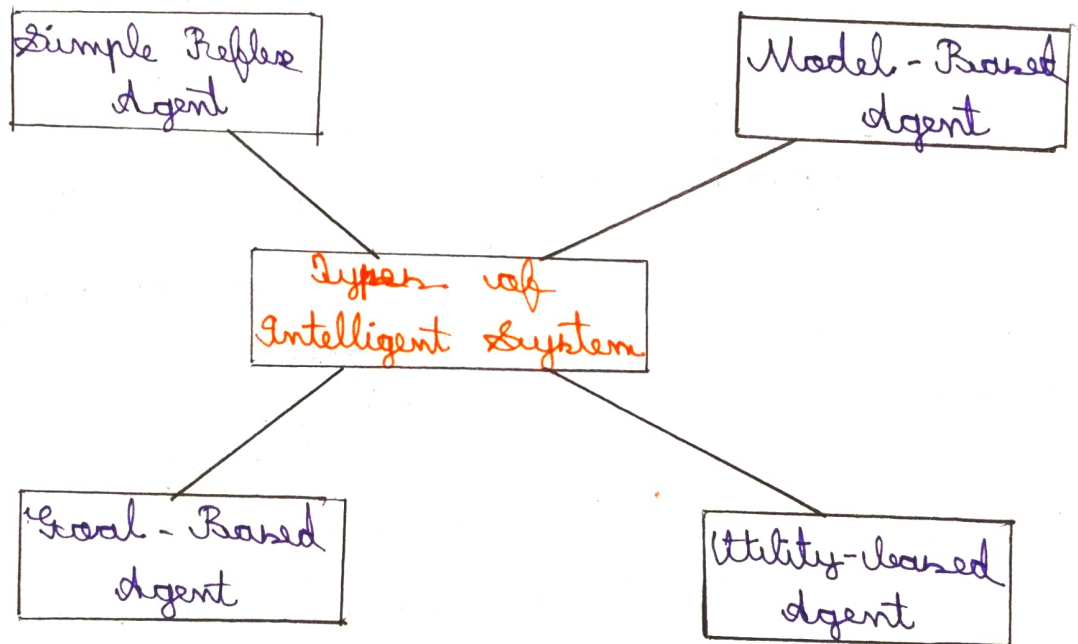
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COURSE CODE : CSA1704  
COURSE NAME: ARTIFICIAL  
INTELLIGENCE

## 1. Smart Home Automation System:

### Introduction:

Smart Home Automation System controls lighting, temperature & security by using sensors such as motion detection, temperature changes & time of day.



### 1. Simple Reflex Agent:

→ Act only on the current input using predefined rules.

→ Eg: Turn on lights when motion is detected.

→ Advantages: Fast & Simple

→ Limitation: Cannot remember past events or learn user preferences.

## 2. Model-Based Agent:

→ Maintains an internal model of the environment.

→ Eg: Knows whether someone is already in the room before turning on the lights.

Advantages: Works well in partially observable environment.

Limitations: Requires more memory & processing.

## 3. Goal-Based Agent:

→ Makes decision to achieve a specific goal.

→ Eg: Maintains room temperature at  $24^{\circ}\text{C}$  while reducing energy usage.

Advantages: Flexible & Intelligent planning.

Limitations: Decision making takes more time.

## 4. Utility-Based Agent:

→ Choose the action with highest overall benefit.

→ Eg: Balance comfort, security & electricity consumption.

→ Advantages: Gives the best overall performance.

→ Limitations: Utility calculation can be complex.

## Best Approach:

A hybrid system contains both goal based & utility based agents will be more effective.

→ Model based understand the home environment.

Utility based agent selects action that maximize comfort, security & energy efficiency.

## 2. Robot in an Unknown Maze:

### Introduction:

Robot in an unknown maze must find the exit without knowing the map. Search Algorithm like Uniform Cost Search (UCS) & Iterative Deepening Search (IDS) helps robot to explore the maze.

### Uniform Cost Search (UCS):

- Expands the path with the lowest cumulative cost.
- Finds the optimal path when movement cost differ.

### Advantage:

- Complete & Optimal.
- Suitable for different path costs.

### Disadvantage:

- High memory usage.
- Slower for large search space.

Complexity:

→ Time: Exponential (depends on branching factor),

→ Space: Exponential.

## 2. Iterative Deepening Search (IDS)

Repetitively, performs depth-limited search with increase depth.

Advantages:

Very low memory usage.

Complete for finite search spaces.

Disadvantages:

Repeats node expansion.

Not optimal when path costs vary.

Complexity:

Time:  $O(b^d)$

Space:  $O(bd)$

## Relation to Intelligent Agents:

### 1. Simple Reflex agent:

Cannot solve an unknown maze b.c it has no memory.

### 2. Model-Based agent:

Stores explored paths & avoids revisiting locations.

### 3. Goal-Based agent:

Searches for the exit using search algorithms.

#### 4. Utility - Based agent:

Selects the safest or least costly path.

#### Best Combination:

a Goal - Based agent with UCS is the most effective. b.c.:

- It plans towards the goal (exit).
- Find the lowest-cost path.
- Makes intelligent decisions in unknown environments.

#### Conclusion:

For the Unknown maze, a Goal - Based agent using UCS provides the most reliable & optimal solution, while IDS is useful when memory is limited.